

CIUDAD REAL, Spain

WMO: 083480

Lat: 38.9892N

Lon: 3.9203W

Elev: 629

StdP: 93.99

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.9	(c) -1.5	(d) -7.9	(e) 2.1	(f) 5.1	(g) -6.0	(h) 2.4	(i) 4.9	(j) 10.3	(k) 10.0	(l) 8.4	(m) 8.4	(n) 0.6	(o) 120	(p) 0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.6	(c) 37.8	(d) 19.8	(e) 36.3	(f) 19.5	(g) 34.9	(h) 19.0	(i) 21.4	(j) 34.5	(k) 20.6	(l) 33.5	(m) 19.9	(n) 32.3	(o) 2.7	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.3	25.2	16.3	12.5	24.4	15.6	11.9	23.8	65.0	34.4	62.0	33.6	59.5	32.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.8	(b) 6.5	(c) 5.5	(e) -5.7	(f) 40.6	(g) 2.2	(h) 1.3	(i) -7.2	(j) 41.5	(k) -8.5	(l) 42.3	(m) -9.7	(n) 43.0	(o) -11.3	(p) 44.0		
(a) 7.8	(b) 6.5	(c) 5.5	(e) -6.3	(f) 23.1	(g) 2.1	(h) 1.6	(i) -7.7	(j) 24.3	(k) -8.9	(l) 25.2	(m) -10.1	(n) 26.2	(o) -11.6	(p) 27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	6.1	7.9	11.0	13.6	18.1	23.8	27.1	26.2	21.1	16.3	10.2	6.9
	DBStd	7.87	2.87	2.89	3.09	3.27	3.78	3.55	2.75	2.51	3.23	2.82	3.09	3.00
	HDD10.0	363	123	68	26	7	1	0	0	0	0	1	35	103
	HDD18.3	1781	378	292	228	146	51	3	0	0	10	75	244	354
	CDD10.0	2458	3	9	57	114	252	414	529	502	333	196	41	7
	CDD18.3	835	0	0	0	3	44	168	271	244	94	12	0	0
	CDH23.3	10753	0	0	7	82	574	2111	3673	3144	1005	156	1	0
CDH26.7	5501	0	0	0	9	201	1073	2096	1701	390	31	0	0	
Wind	WSAvg	2.1	1.9	2.4	2.5	2.4	2.2	2.2	2.1	2.1	1.9	1.7	1.8	1.8
Precipitation	PrecAvg	428	46	45	40	46	40	28	8	6	25	48	52	55
	PrecMax	708	177	156	128	119	144	133	71	25	104	138	122	220
	PrecMin	248	0	0	0	4	3	0	0	0	0	0	1	0
	PrecStd	109	39	41	33	29	29	30	16	7	24	38	32	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.5	24.2	27.9	33.6	38.5	39.9	39.0	36.2	29.6	21.8	16.3
		MCWB	10.2	11.1	13.3	15.2	17.9	19.7	19.8	20.3	19.0	16.0	13.5	11.0
	2%	DB	14.0	17.1	22.2	25.8	31.0	36.1	38.1	37.1	32.9	27.1	19.1	14.5
		MCWB	9.5	10.3	12.2	14.4	17.2	19.2	19.9	19.9	18.5	15.3	12.8	11.5
	5%	DB	12.7	15.2	20.4	23.7	28.8	34.2	36.8	35.7	30.7	24.9	17.2	13.2
		MCWB	9.4	9.7	11.6	13.4	16.2	18.5	19.8	19.5	17.8	15.3	12.4	10.8
	10%	DB	11.3	13.5	18.2	21.3	26.6	32.6	35.2	34.2	28.9	22.8	15.7	12.0
		MCWB	8.6	9.3	10.7	12.4	15.4	17.9	19.0	19.0	17.3	14.9	11.9	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.7	14.6	16.6	19.6	21.3	22.7	23.0	21.2	18.2	16.0	13.4
		MCDB	14.2	16.5	22.6	26.0	31.0	34.4	36.7	34.9	31.2	24.9	18.6	14.5
	2%	WB	11.0	11.6	13.1	15.0	18.1	20.3	21.4	21.5	19.8	17.2	14.6	12.2
		MCDB	12.8	15.2	20.1	23.6	28.0	33.4	35.5	34.1	29.6	23.2	17.2	13.6
	5%	WB	10.0	10.6	12.2	14.1	16.9	19.5	20.5	20.7	18.8	16.4	13.4	11.1
		MCDB	11.8	14.0	18.6	22.0	26.5	31.9	34.3	33.1	28.1	21.8	15.9	12.8
	10%	WB	9.0	9.7	11.3	13.1	16.0	18.7	19.8	19.8	18.0	15.7	12.4	10.0
		MCDB	10.9	12.8	16.9	20.0	25.0	30.1	33.0	31.7	26.8	20.6	14.9	11.9
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.6	12.4	12.0	12.5	14.1	14.6	14.7	12.9	11.5	9.8	8.6
		MCDBR	12.2	14.5	16.9	16.6	17.1	16.8	16.8	17.0	16.0	15.7	12.7	10.0
		MCWBR	8.2	8.3	8.3	7.3	6.7	5.9	6.2	6.4	6.4	6.9	7.4	6.8
	5% WB	MCDBR	8.6	10.9	14.4	14.6	15.0	15.1	15.2	15.4	13.9	11.9	8.9	7.8
		MCWBR	6.5	6.8	7.4	6.8	6.3	5.6	5.9	6.2	5.9	5.6	5.8	5.9
Clear-Sky Solar Irradiance	taub	0.289	0.308	0.343	0.347	0.362	0.366	0.348	0.357	0.350	0.335	0.306	0.291	
	taud	2.507	2.466	2.379	2.377	2.364	2.375	2.404	2.385	2.418	2.470	2.510	2.516	
	Ebn at Noon	895	920	915	929	916	909	923	908	895	875	865	865	
	Edh at Noon	76	91	110	118	122	121	116	115	105	90	76	71	
All-Sky Solar Radiation	RadAvg	2.31	3.20	4.40	5.56	6.65	7.63	7.90	6.89	5.30	3.69	2.57	2.09	
	RadStd	0.33	0.42	0.49	0.46	0.54	0.30	0.19	0.21	0.30	0.31	0.30	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	+0.57	-0.65	-0.55	N/A	N/A	N/A	+69

Nomenclature: See separate page